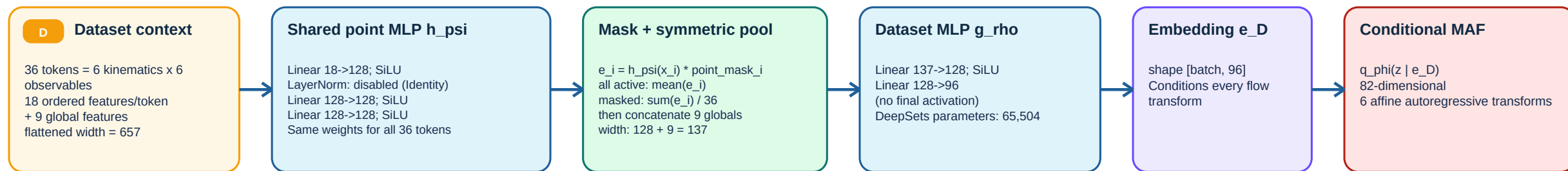


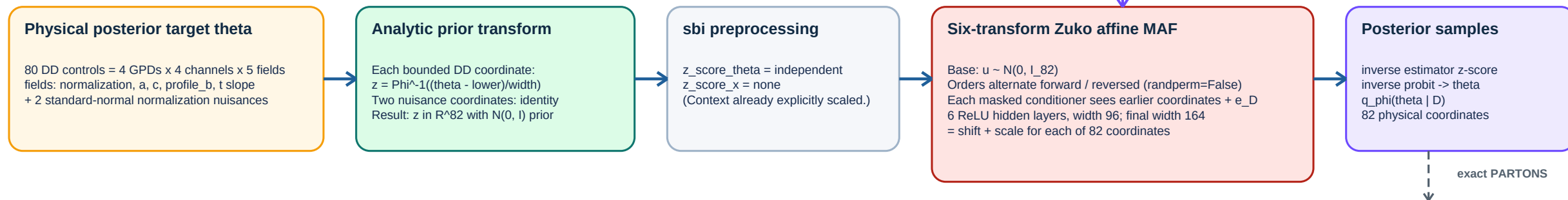
DD + DeepSets + conditional MAF

Registered family: dd_deepsets_maf | current production training path | PARTONS remains the sole forward-physics backend

A. Observation encoder (permutation invariant)



B. Exact target transform and density estimator



Frozen input contract

18 point features (ordered): $x_B/0.3$; $-t/0.2$; Q^2/Q_0^2 ; $E_{\text{beam}}/12$; $\sin(\phi)$; $\cos(\phi)$; six observable one-hot bits; observed value; marginal sigma; symmetric $C^{(-1/2)y/10}$; global-normalization response; LU-normalization response; point_mask. Nine globals (ordered): PARTONS-backend flag; $Q_0^2/(1 \text{ GeV}^2)$; LO-coefficient flag; configured-evolution flag; native datum-Q2 evolution flag; full-covariance flag; four-twist-2-GPD flag; native-six-observable-route flag; model_mask.

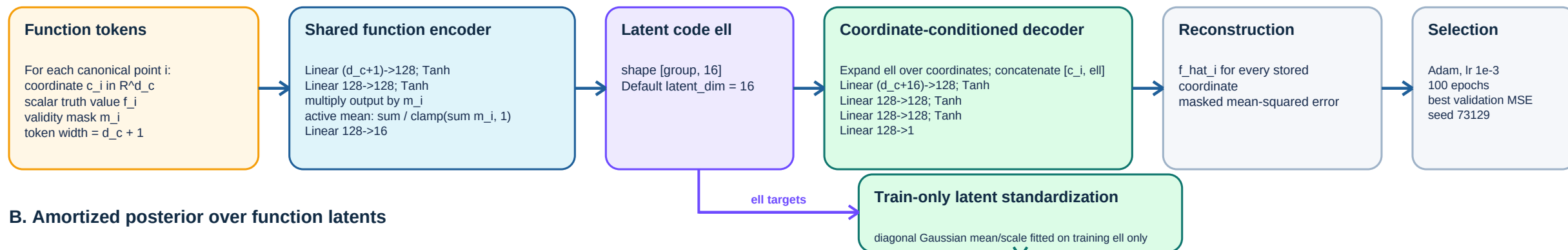
Interpretation and boundary

The network learns a normalized joint posterior, not 82 independent regressions. Dashed downstream physics is intentionally outside the neural network: posterior samples are reevaluated by the exact native PARTONS bridge for CFFs, observables, and GPD curves.

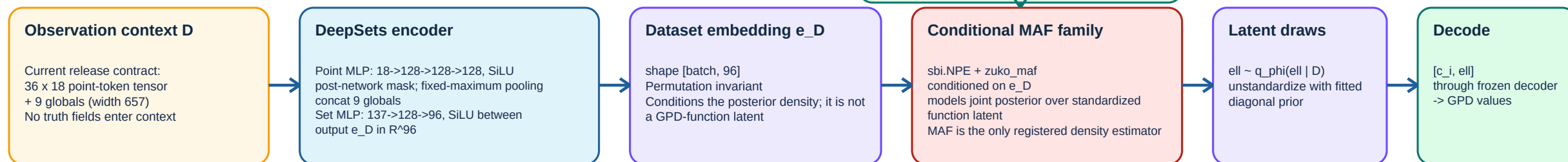
NNGPD-inspired latent function + DeepSets + conditional MAF

Registered family: neural_gpd_deepsets_maf | function autoencoder v1 | latent-function closure representation

A. Function autoencoder (trained only from stored canonical GPD truth)



B. Amortized posterior over function latents



Exact scope and safeguards

Identifier: permutation_invariant_function_autoencoder_v1; decoder: coordinate_conditioned_mlp_v1. Default d_c is artifact-defined (the checkpoint records coordinate_width), so it is intentionally symbolic here. The support guard rejects unsupported coordinates, and the published decoder contract uses extrapolation_policy="forbidden". Its fitted latent prior uses training groups only, and train / validation / locked-holdout roles must be nonempty and disjoint.

Implementation status (current worktree)

The family and function autoencoder/model-view publisher are implemented and registered. The ordinary workflows/pseudodata.py train_model() path still loads theta_latent and writes a dd_deepsets_maf result bundle. Therefore this page describes the registered neural-GPD architecture without claiming that the standard CLI currently trains it end-to-end. The neural model-view publisher does not yet record concrete MAF widths/transform count.